

230905232
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CSE C2 Roll No. 37

WPL Week 4

Home Assignments

1. Write a python program to select smallest element from a list in an expected linear time.

Code:

```
arr = list(map(int, input("Enter elements of the list: ").split()))
smallest = float('inf')
for i in arr:
    if smallest > i: smallest = i
print("smallest element is", smallest)
```

output:

```
aditya-mathpal@ThinkPad-E14-Gen-5:~/codes/230905232/WP-Lab/lab04-python-basics/home$ python3 1.py
Enter elements of the list: 7 2 9 5 1 6
smallest element is 1
```

2. Write a python program to implement bubble sort.

Code:

```
def bubble_sort(arr):
    n = len(arr)
    swapped = False
    for i in range(n-1):
        swapped = False
        for j in range(n-i-1):
            if arr[j] > arr[j+1]:
                arr[j], arr[j+1] = arr[j+1], arr[j]
                swapped = True
        if not swapped: return

arr = list(map(int, input("Enter elements of the list: ").split()))
bubble_sort(arr)
print("sorted list:", arr)
```

output:

```
aditya-mathpal@ThinkPad-E14-Gen-5:~/codes/230905232/WP-Lab/lab04-python-basics/home$ python3 2.py
Enter elements of the list: 7 6 5 4 3 2 1
sorted list: [1, 2, 3, 4, 5, 6, 7]
```

3. Write a python program to multiply two matrices

code:

```
import numpy as np
```

```
def mat_product(matA, matB):
    m, n = len(matA), len(matB)
    res = [[0]*m for i in range(m)]
    for i in range(m):
        for j in range(m):
            for k in range(n):
                res[i][j] += matA[i][k] * matB[k][j]
    return res
```

```
def read_mat_input(mat):
    m, n = len(mat), len(mat[0])
    for i in range(m):
        mat[i] = list(map(int, input().split()))
```

```
m, n = list(map(int, input("Enter dimensions of matrix A: ").split()))
o, p = list(map(int, input("Enter dimensions of matrix B: ").split()))
if m != p or n != o:
    print("Enter valid dimensions")
    raise SystemExit(1)
```

```
matA = [[0]*n for i in range(m)]
matB = [[0]*m for i in range(n)]
```

```
print("Enter elements of matrix A:")
read_mat_input(matA)
print("Enter elements of matrix B:")
read_mat_input(matB)
```

```
print("matA x matB =\n", np.array(mat_product(matA, matB)))
```

output:

```
(venv) aditya-mathpal@ThinkPad-E14-Gen-5:~/codes/230905232/WP-Lab/lab04-python-basics/home$ python3 3.py
Enter dimensions of matrix A: 2 3
Enter dimensions of matrix B: 3 2
Enter elements of matrix A:
1 2 3
4 5 6
Enter elements of matrix B:
10 11
20 21
30 31
matA x matB =
[[140 146]
 [320 335]]
```

4. Write a Python class to find validity of a string of parentheses, '(', ')', '{', '}', '[', and ']'. These brackets must be close in the correct order, for example "()" and "()[]{}" are valid but "(", "{[]}" and "{{{" are invalid.

Code:

```
def verify_string(s):
    stack = []
    dict = {')': '(', '}': '{', ']': '['}
    for i in s:
        if i in '([{':
            stack.append(i)
        elif dict[i] == stack[-1]:
            stack.pop()
        else:
            return 0
    return not stack
```

```
s = input("Enter string: ")
if verify_string(s):
    print("Valid string")
else:
    print("Invalid string")
```

output:

```
aditya-mathpal@ThinkPad-E14-Gen-5:~/codes/230905232/WP-Lab/lab04-python-basics/home$ python3 4.py
Enter string: ()[]{}
Valid string
aditya-mathpal@ThinkPad-E14-Gen-5:~/codes/230905232/WP-Lab/lab04-python-basics/home$ python3 4.py
Enter string: {{{
Invalid string
```

5. Write a Python class to reverse a string word by word.

Code:

```
class Reverse:
    def __init__(self, s):
        self.s = s

    def rev(self):
        words = self.s.split()
        res = ' '.join(words[::-1])
        return res

words = input("Enter string: ")
obj = Reverse(words)
output = obj.rev()
print("Words reversed:", output)
```

output:

```
aditya-mathpal@ThinkPad-E14-Gen-5:~/codes/230905232/WP-Lab/lab04-python-basics/home$ python3 5.py
Enter string: this is a test
Words reversed: test a is this
```

6. Write a Python class named Circle constructed by a radius and two methods which will compute the area and the perimeter of a circle.

Code:

```
import math
```

```
class Circle:
```

```
    def __init__(self, radius):
        self.radius = radius
```

```
    def area(self):
        return math.pi*self.radius**2
```

```
    def perimeter(self):
        return 2*math.pi*self.radius
```

```
r = float(input("Enter radius of circle: "))
```

```
obj = Circle(r)
```

```
print(f"Area of circle is {round(obj.area(), 4)}, perimeter is {round(obj.perimeter(), 4)}")
```

output:

```
aditya-mathpal@ThinkPad-E14-Gen-5:~/codes/230905232/WP-Lab/lab04-python-basics/home$ python3 6.py
Enter radius of circle: 4
Area of circle is 50.2655, perimeter is 25.1327
```